Status	Finished
Started	Tuesday, 20 October 2026, 11:26 AM
Completed	Tuesday, 20 October 2026, 11:31 AM
Duration	5 mins 7 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Bubble Sort
b. Linear Search
c. Quick Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Bubble Sort
b. Quick Sort
c. Merge Sort
d. Heap Sort

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are merged without sorting
b. Each part is sorted independently
c. Each part is searched for a specific element
d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **4**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are compared element by element
c. They are combined to form a single sorted list
d. They are split again into smaller sublists

The correct answer is: They are combined to form a single sorted list

Question **5**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Binary Search
c. Merge Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Quick sort
b. Merge sort
c. Heap sort
d. Selection sort

The correct answer is:
Merge sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It uses the heap data structure
b. It divides the list into sublists
c. It recursively sorts subproblems
d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **8**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Using the divide-and-conquer approach
b. Sorting data in a non-sequential manner
c. Being the most efficient sorting algorithm
d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **9**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By using the bubble-up method
b. By sorting data in a single pass
c. By comparing elements sequentially
d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **10**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Complexity
- b. Merging
- c. Sorting
- d. Searching

The correct answer is:
Complexity

Question **11**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Searching for data in a list
- b. Inserting data into a list
- c. Deleting data from a list
- d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format

Question **12**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Makes code execution slower
- c. Makes data harder to manage
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **13**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Heap sort

b. Quick sort

c. Bubble sort

d. Insertion sort

The correct answer is:
Insertion sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items

b. Sorting is rarely used in programming

c. Sorting decreases the efficiency of selection operations

d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **15**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. arrange()

b. sorted()

c. sort()

d. order()

The correct answer is: sorted()